

PUBPOL 5390 Homework 1

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1 Predicting Birthweight

1.1 Data Cleaning

To clean the data, I read the provided documentation that described the type and missingness coding for each variable. Then, I went through the data and renamed variables to be more descriptive and filtered out missing data based on the coding scheme (e.g., removing 9999 observations in numeric variables and “NA” observations in character variables).

1.2 Modeling Choices

1.3 In and Out-of-Sample Goodness of Fit

Overall, the models do a reasonable job at predicting birthweight. The training birthweight SD is 585.80 and the testing is 595.08, which are very similar. For the in-sample OLS, the first feature predicts the birthweight

Table 1: Summary of Prediction Performance Across Models

Sample	SD Birthweight	N Features Set 1	N Features Set 2	N Features Set 3	Mod	RMSE	R-Squared
Train	585.80	9.00	37.00	164.00	Set 1	481.36	0.32
Train	585.80	9.00	37.00	164.00	Set 2	465.61	0.37
Train	585.80	9.00	37.00	164.00	Set 3	455.67	0.40
Test	595.08	9.00	37.00	164.00	Set 1	492.38	0.32
Test	595.08	9.00	37.00	164.00	Set 2	476.18	0.36
Test	595.08	9.00	37.00	164.00	Set 3	464.21	0.39

about 100grams better than the SD in both the train and test sample. As we add more features, we get minor gains in predictive performance, but this is only about 30 grams for the most “overfit” model. Across models, the in-sample R-squared is slightly higher.

Table 1 describes the in and out-of-sample goodness of fit statistics for each of the three feature sets used. In the first model, the in-sample RMSE approximates the out-of-sample RMSE quite well. The difference is within about 10 units. In the second model, the difference is also similar. In the third model, we see the same pattern. Each model’s in-sample fit understates the RMSE out-of-sample RMSE by 10 or so points.

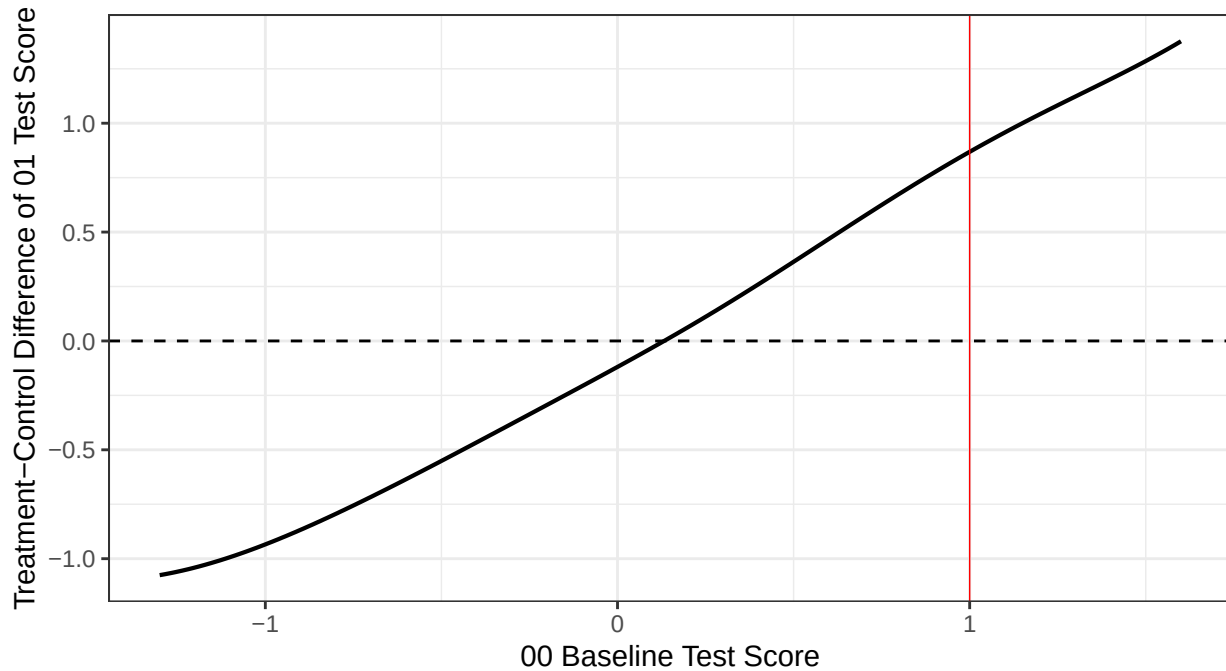
A key lesson from this exercise is that conventional OLS methods tend to overstate the predictive performance of models on in-sample data. This is because OLS is minimizing with respect to the sample. It is therefore necessary to evaluate machine learning tools on testing samples to understand how they fit data they have never seen. In general, as we add predictors, we can enrich our predictions, but this may result in overfitting which is problematic for making good predictions.

2 Problem 2

2.1 i-iii

(2001) Test Score Impacts by Baseline (2000) Test Score Control Girls

Nonparametric Fan Locally Weighted Regression



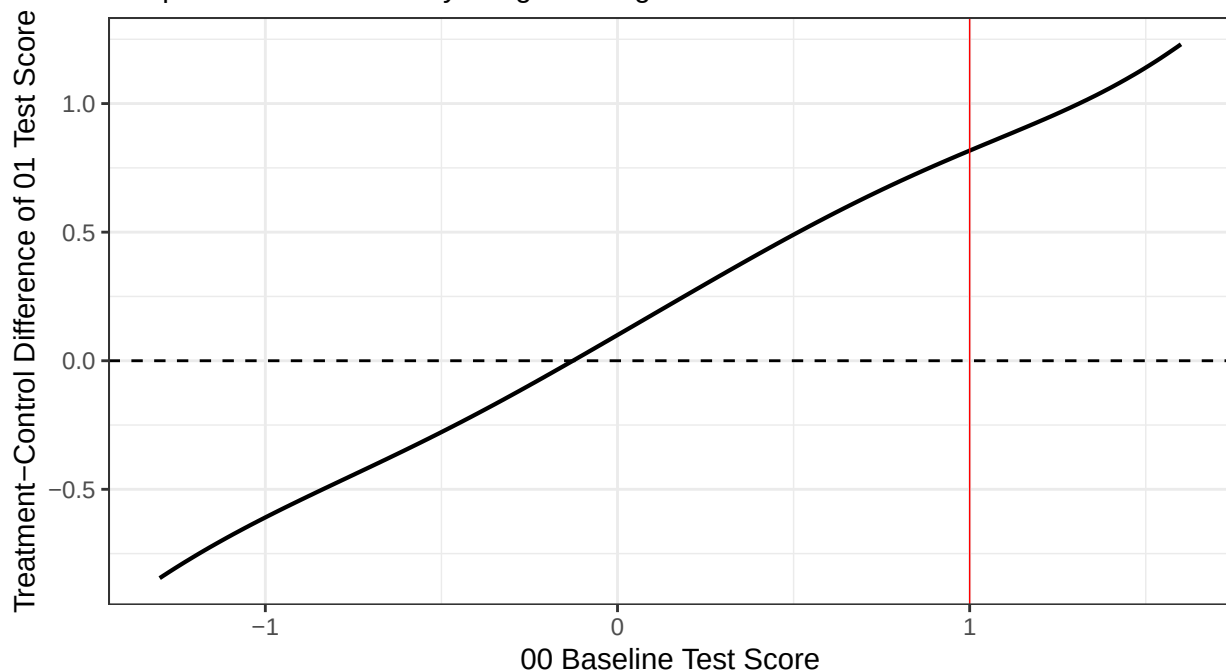
Dashed line represents 0 standardized test score.

Horizontal line represents minimum winning score in 2001.

Local linear regressions use Epanechnikov kernel with .4 bandwidth.

(2001) Test Score Impacts by Baseline (2000) Test Score Treated Girls

Nonparametric Fan Locally Weighted Regression



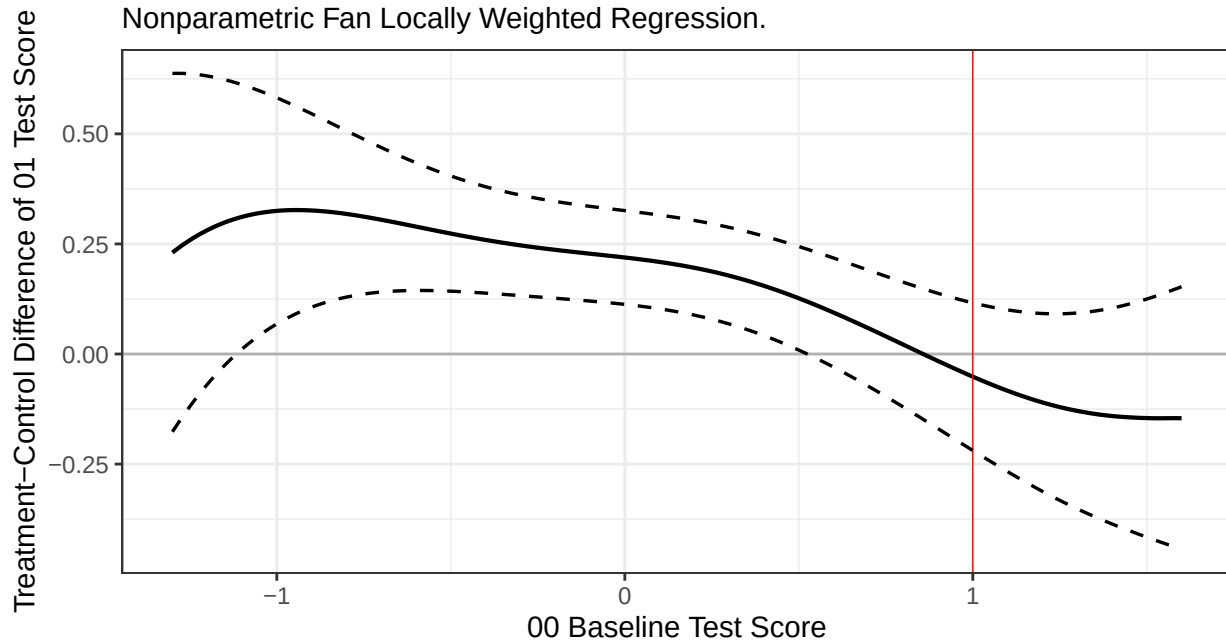
Dashed line represents 0 standardized test score.

Horizontal line represents minimum winning score in 2001.

Local linear regressions use Epanechnikov kernel with .4 bandwidth.

Figure 3a: Year 1 (2001) Test Score Impacts by Baseline (2000) Test Score Difference Between Program and Comparison Schools, Cohort 1

Nonparametric Fan Locally Weighted Regression.



Dashed line represents 0 standardized test score.

Horizontal line represents minimum winning score in 2001.

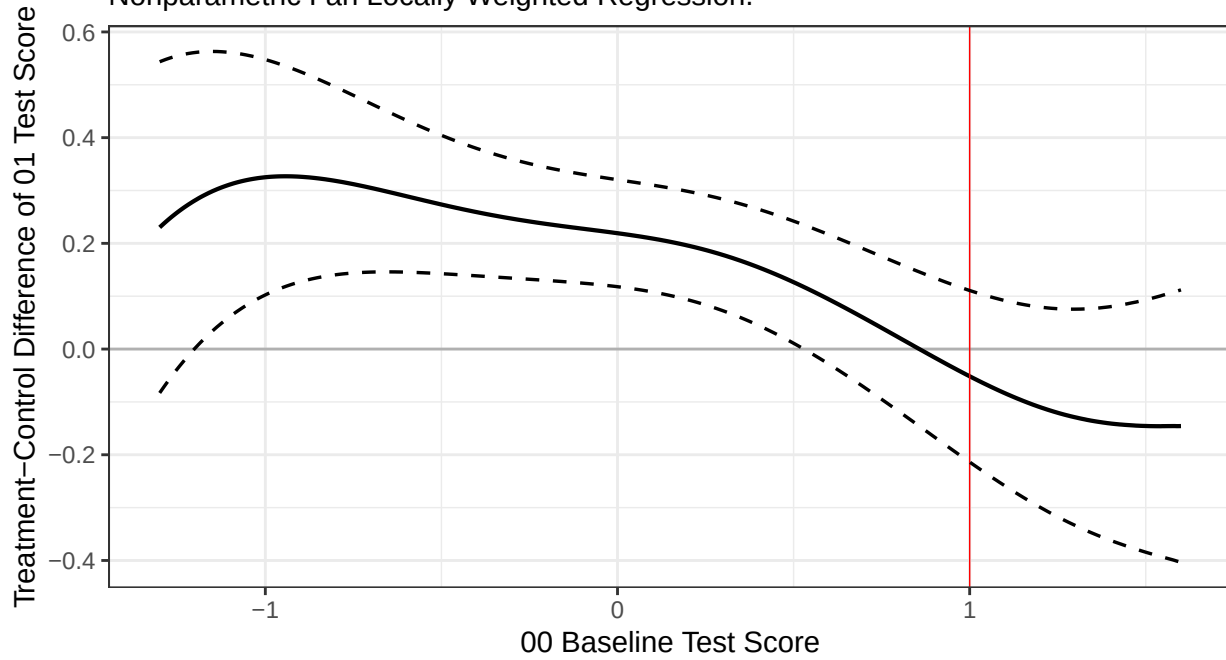
Local linear regressions use Epanechnikov kernel with .7 bandwidth. Bootstrapped CIs based on 50 replications.

Dashed lines represent bootstrapped 95% confidence intervals of the treatment control difference.

2.2 iv-vi

Figure 3a: (Large Bandwidth)

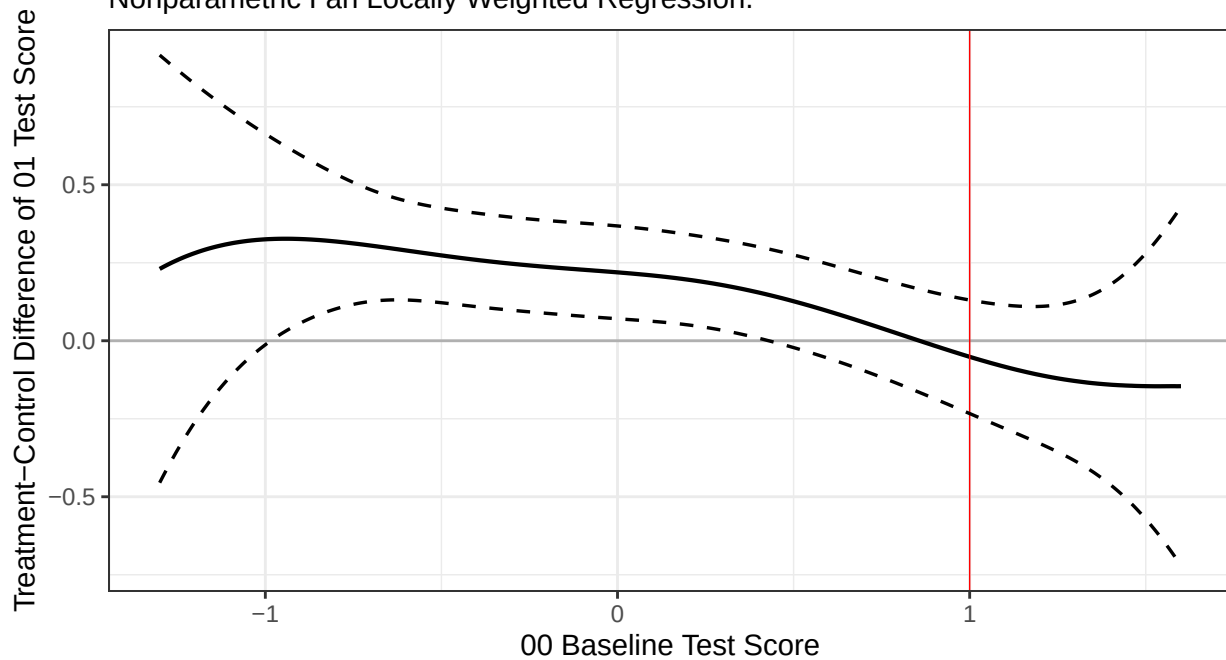
Nonparametric Fan Locally Weighted Regression.



Dashed line represents 0 standardized test score.
Horizontal line represents minimum winning score in 2001.
Local linear regressions use Epanechnikov kernel with 1 bandwidth. Bootstrapped CIs based on 50 replications.
Dashed lines represent bootstrapped 95% confidence intervals of the treatment control difference.

Figure 3a: (Small Bandwidth)

Nonparametric Fan Locally Weighted Regression.



Dashed line represents 0 standardized test score.

Horizontal line represents minimum winning score in 2001.

Local linear regressions use Epanechnikov kernel with .3 bandwidth. Bootstrapped CIs based on 50 replications.

Dashed lines represent bootstrapped 95% confidence intervals of the treatment control difference.

Replicating Figure 3a with a larger bandwidth reveals much the same relationship as with the Kremer et al. bandwidth. There appears to be a small decrease in the treatment-control difference around the 1 baseline score in 2000, but they appear generally the same.

Replicating 3a with the smaller bandwidth changes the interpretation somewhat. With a bandwidth of .3, the line appears almost completely flat and the uncertainty around the ends of the distribution are much wider. With a smaller bandwidth, the results suggest that there is not much difference in the effect across baseline test scores.

Consistent with Kremer et al., I prefer the .7 bandwidth estimate. This estimate suggests that the test score treatment effects were the strongest for girls who had test scores between -1 and .8 in 2000, with those on the ends of the distribution having effects indistinguishable from 0.

Table 2: Replication of Mullinathan and Spiess Table 1

method	IS RMSE	IS R2	IS SD(Ln(Value))	OOS RMSE	oosse	OOS R2	OOS SD(Ln(Value))
OLS	0.843	0.416	4.921	0.859	0.035	0.395	5.066

Table 3: Replication of Mullinathan and Spiess Table 1 (KS)

method	IS RMSE	IS R2	IS SD(Ln(Value))	OOS RMSE	oosse	OOS R2	OOS SD(Ln(Value))
OLS	0.81	0.46	4.92	0.83	0.04	0.43	5.11

3 Problem 3

3.1 Data Cleaning

Using the PUF flat file, I am not able to get their sample size exactly. This is because it appears that they are merging it with a “metropolitan” dataset that is not apart of the national data. However, their code does not detail this merging step beforehand, so it is not possible to reproduce it directly. I end up with an analytic sample size of 30,610. To make this fully reproducible, there needs to be a greater discussion of where the metropolitan data is coming from and how the national and metropolitan data were merged.

3.2 b-c Table Replication

To interpret the estimate of the RMSE in comparison to the $\log(\text{house price})$, we would say that the model improves at predicting house price by 4.2 log units compared to an estimate to using the mean value.

3.3 d-e Kitchen Sink Regression

In the updated “kitchen sink model,” I employ an extensive set of factor variable versions and include deciles of lot size to account for non-linearities related to the size of dwellings. These transformations improve on the in and out-of-sample R2 and RMSE above what is presented in Mullinathan and Spiess. I improve the in-sample R2 by .04 and the out of sample by .03.

Across quintiles, my OLS model improves over the MS OLS model in terms of R2, but the change is uneven

Table 4: Relative Improvement in KS OLS
over MS OLS

P1	P2	P3	P4	P5
-0.09	-0.29	3.40	-0.15	-0.14

across price quintiles. In all quintiles, but the 3rd my predictions become worse. However, in the 3rd quintile, the predictions improve substantially.

3.4 f

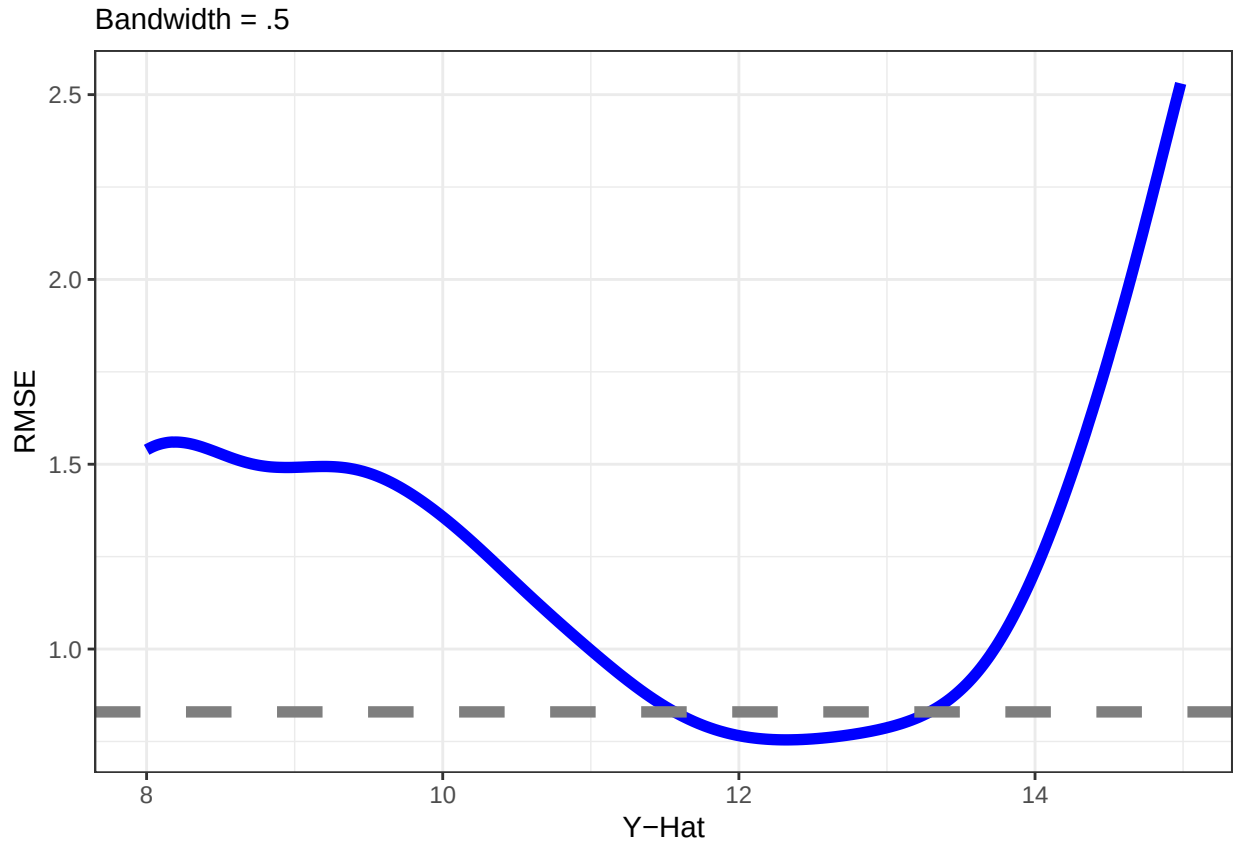


Figure 1: Fan Regression Estimates of RMSE by Y-Hat

Across the distribution of Y-hat values, we can see that the, compared to the global RMSE, the model predicts better for values of y-hat between 8-14.99. This is because most of the data exist in this space owing to the skewed distribution.

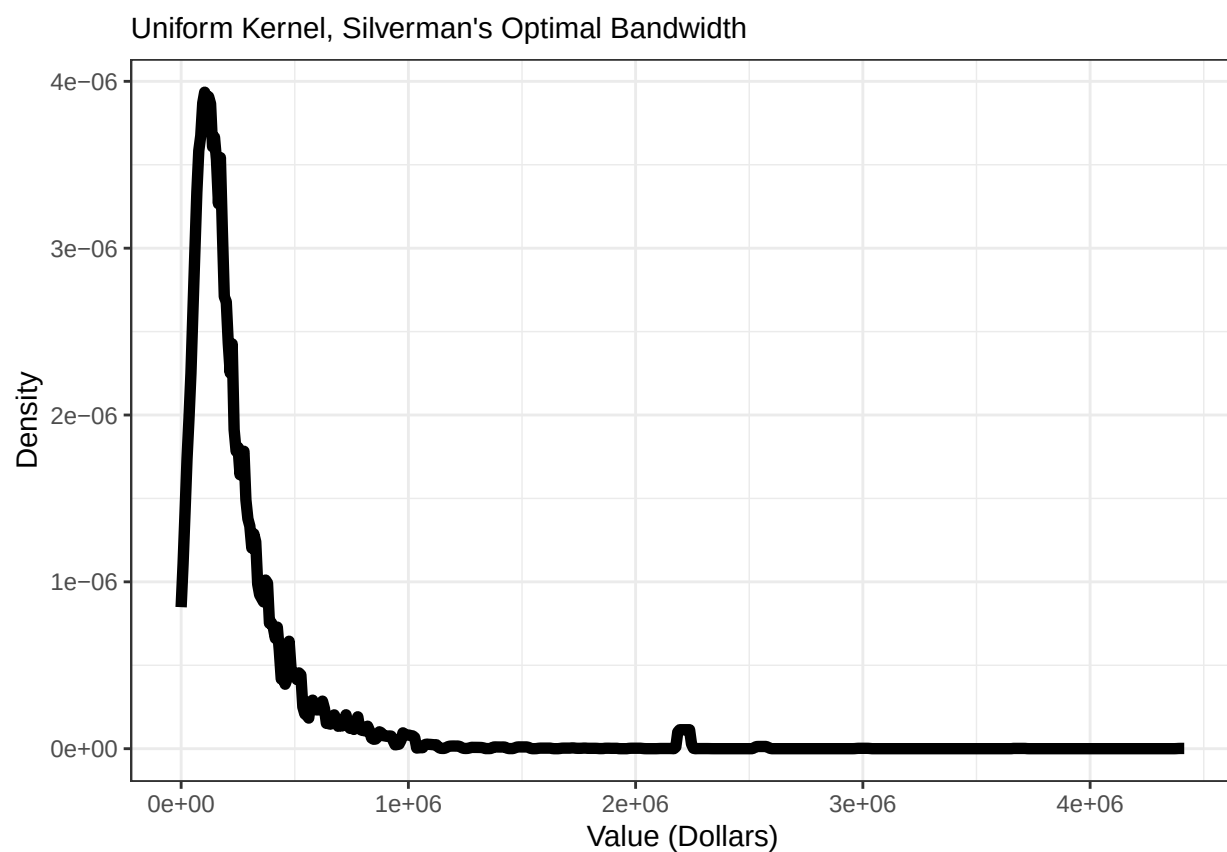


Figure 2: Density of House Prices

Table 5: Summary Statistics of VALUE

Mean	SD	Min	Max	Skewness
240 359.98	270 043.31	1.00	4 414 135.00	4.65

Table 6: Comparison of Raw House Values and Summary Statistics Across Models

Model	SD Value	RMSE	R2
MS OLS	270 644.43	239 569.54	0.22
KS OLS	270 644.43	232 553.99	0.26

Table 7: Replication of MS Model (Raw Values)

method	IS RMSE	IS R2	IS SD(Value)	OOS RMSE	oosse	OOS R2	OOS SD(Value)
OLS	224 913.83	0.30	184 026.91	231 729.99	2 309 829 331.85	0.27	190 139.50

3.5 g

Based on the kernel density and the summary statistics, house prices appear very skewed to the right.

3.6 h

If we transform $\log(\text{house prices})$ back to their raw values, we see that the models decrease in predictive performance substantially and only offer small improvements over using the mean values. The OOS R2 is also significantly reduced (by about half the original value).

Recomputing both the MS and KS models, we see that the performance on level house prices improves compared to the exponentiated values. However these improvements are not to the level of using the logged house price variable.

Table 8: KS Model (Raw Values)

method	IS RMSE	IS R2	IS SD(Value)	OOS RMSE	oosse	OOS R2	OOS SD(Value)
OLS	216 402.99	0.35	178 626.65	224 392.51	2 229 312 987.62	0.31	185 778.71